

Why Python For Data Science

Objectives

- What is the Python Programming Language
- Why use Python for data science
- What are Jupyter Notebooks

What is the Python Programming Language

Python was designed by Guido van Rossum and first became available for use in 1991. Python is a great first programming language because its syntax (vocabulary of the language) is not complicated when compared to many other languages. This will allow you to focus on learning programming and data science concepts rather than complicated language.

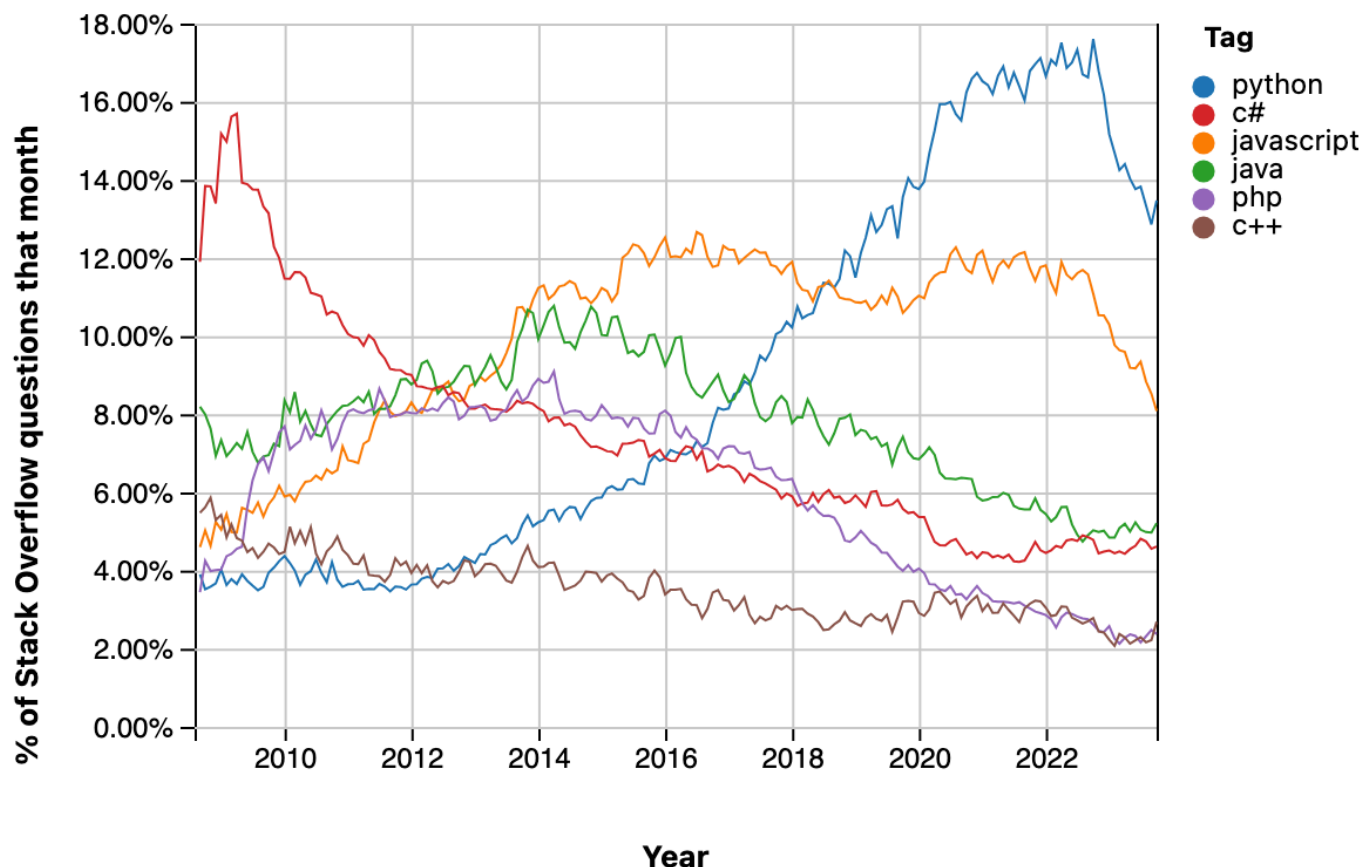
One important thing to remember is that Python's simplicity relies upon proper indentation, which you'll find helps you to write more clean and readable code. If you've dabbled with HTML or other languages where coding style is emphasized, you know the importance of indentation already, but you'll soon learn why Python requires it.

Why use Python for Data Science

The simplest reason is that Python is very popular programming language. The image below shows the percentage of Stack Overflow questions over time ([source](#) 

([https://insights.stackoverflow.com/trends?](https://insights.stackoverflow.com/trends?tags=python%2Cjavascript%2Cjava%2Cc%23%2Cphp%2Cc%2B%2B&utm_source=so-owned&utm_medium=blog&utm_campaign=gen-blog&utm_content=blog-link&utm_term=incredible-growth-python)

[tags=python%2Cjavascript%2Cjava%2Cc%23%2Cphp%2Cc%2B%2B&utm_source=so-owned&utm_medium=blog&utm_campaign=gen-blog&utm_content=blog-link&utm_term=incredible-growth-python](https://insights.stackoverflow.com/trends?tags=python%2Cjavascript%2Cjava%2Cc%23%2Cphp%2Cc%2B%2B&utm_source=so-owned&utm_medium=blog&utm_campaign=gen-blog&utm_content=blog-link&utm_term=incredible-growth-python)). While this is **not necessarily all due to the usage of Python for data science** , it is a big part of the reason why. (<https://stackoverflow.blog/2017/09/06/incredible-growth-python/>),



You may be wondering why should popularity be a reason for using a language. Having a large community of Python users means you are able likely to get help from others when you ask. One consequence of the language being popular is that you will find that there is a Python library you can import to help accomplish your analytical or machine learning goals without writing all your code from scratch. This course utilizes many Python libraries including NumPy, SciPy, pandas, matplotlib, seaborn, and scikit-learn as well as the deep learning framework PyTorch.

What are Jupyter Notebooks

In this course, we will use Colab Notebooks which are essentially Jupyter Notebooks freely hosted by Google so you don't have to install anything on your computer. Jupyter Notebooks contain both code and rich text elements, such as figures, links, and equations. The reason why it is important to not just have code but also text explaining your code and your results is that a lot of data science is about communicating what the data means to people. The image below is a Jupyter Notebook. If you want to learn how to use a Jupyter Notebook, there is an excellent resource from [Corey Schafer on getting started with Jupyter Notebooks](https://youtu.be/HW29067qVWk?si=R0ljv8-SKHjrkDgV) [↗](https://youtu.be/HW29067qVWk?si=R0ljv8-SKHjrkDgV) [. \(https://youtu.be/HW29067qVWk?si=R0ljv8-SKHjrkDgV\)](https://youtu.be/HW29067qVWk?si=R0ljv8-SKHjrkDgV).

What is `train_test_split`

1. Split the dataset into two pieces: a **training set** and a **testing set**. Typically, about 75% of the data goes to your training set and 25% goes to your test set.
2. Train the model on the **training set**.
3. Test the model on the **testing set** and evaluate the performance

Import Libraries

```
In [1]: %matplotlib inline

import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split

from sklearn.linear_model import LinearRegression
```

Load the Dataset

The code below loads and displays the Boston dataset.

```
In [2]: df = pd.read_csv("https://raw.githubusercontent.com/mGalarnyk/Tutorial_Data/master/Boston_Housing/bo
```

```
In [3]: df.head()
```

```
Out[3]:
```

	CRIM	ZN	INDUS	CHAS	NOX	RM	AGE	DIS	RAD	TAX	PTRATIO	B	LSTAT	target
0	0.00632	18.0	2.31	0.0	0.538	6.575	65.2	4.0900	1.0	296.0	15.3	396.90	4.98	24.0
1	0.02731	0.0	7.07	0.0	0.469	6.421	78.9	4.9671	2.0	242.0	17.8	396.90	9.14	21.6
2	0.02729	0.0	7.07	0.0	0.469	7.185	61.1	4.9671	2.0	242.0	17.8	392.83	4.03	34.7
3	0.03237	0.0	2.18	0.0	0.458	6.998	45.8	6.0622	3.0	222.0	18.7	394.63	2.94	33.4
4	0.06905	0.0	2.18	0.0	0.458	7.147	54.2	6.0622	3.0	222.0	18.7	396.90	5.33	36.2